Mukul Rayana

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Technical Skills

- **Languages:** Python, C++, C#, SQL
- **Software Engineering:** Data Structures & Algorithms, Object-Oriented Design, REST APIs, Version Control (Git), CI/CD, Debugging & Testing
- **AI/ML:** PyTorch, Hugging Face Transformers, LangGraph, LangChain, Llama 3.1 (QLoRA/PEFT), Retrieval-Augmented Generation, Reinforcement Learning (PPO, Ray RLlib), scikit-learn, XGBoost
- **Databases & Search:** PostgreSQL, MongoDB, SQLite, pgvector, Pinecone, FAISS, ChromaDB
- **Cloud & DevOps:** AWS (EC2, S3), Azure Container Apps, Docker, Kubernetes, FastAPI, GitHub Actions, MLflow

Education

University of Maryland, College Park Aug 2024 -- May 2026
M.S., Applied Machine Learning *College Park, MD*
Coursework: Algorithms & Data Structures, Computing Systems for ML, Cloud Computing, Optimization, NLP, Computer Vision

SRM Easwari Engineering College Dec 2020 -- Apr 2024
B.Tech., Artificial Intelligence & Data Science *Chennai, India*
Coursework: Programming & Data Structures, Object-Oriented Programming, DBMS (SQL/NoSQL), Deep Learning | Certification: Data Structures & Algorithms Specialization (UC San Diego)

Experience

BeeBox Studios Pvt Ltd | IIT Madras Research Park Apr 2023 -- Apr 2024
AI Research Intern *Chennai, India*
- Built a BERT-based QA system for real-time in-headset lookup over a 100-page XR corpus (**83.2% F1**, 71.8% Exact Match), replacing manual reference entirely; ONNX export + INT8 quantization cut P95 latency **3**× (185 ms → 62 ms) on GPU-constrained hardware.
- Led as the sole ML engineer across three cross-functional teams (UI/UX, WebRTC, ML); contributed to a CI/CD deployment pipeline (Docker, GitHub Actions) supporting **six production releases** with no service interruption, and integrated the Python model with the team's Unity/C# frontend over real-time TCP.

Projects

PharmaChain -- Agentic Cold-Chain Incident Resolution System Apr 2026
- **Won 1st Place ($4K)** at the UMD Smith Agentic AI Challenge, helping design an 8-agent LangGraph system that resolves pharmaceutical cold-chain incidents in 12--15s, cutting a manual multi-hour triage process down to real-time while gating every irreversible action behind human approval.
- Owned the Route and Triage agents and their REST API layer, a 4-provider LLM fallback chain (hot-swap, graceful degradation to deterministic mode), and a bug fix eliminating one wasted LLM call per incident; integrates an Optuna-tuned XGBoost risk engine (**0.94 ROC-AUC**) with SHAP explanations and RAG over 417 regulatory chunks.

Irminsul -- Full-Stack LLM Serving Platform Nov 2025 -- Jan 2026
- Independently built and shipped a full LLM serving platform end-to-end -- fine-tuning, retrieval, safety guardrails, and hosting -- for fast, low-cost, reliably-grounded inference; QLoRA fine-tuned Llama 3.1 8B (3 MLflow-tracked runs), served via Dockerized FastAPI on Hugging Face Spaces with a **6,876-vector** Pinecone index, prompt-injection detection, and domain cosine guardrails.
- Automated the data pipeline end-to-end: GitHub Actions triggers weekly corpus refresh (840 docs) with incremental re-indexing and zero-downtime redeployment with no manual steps; profiled and optimized retrieval latency across the pipeline pre-deployment.

RECON -- Multi-Agent Literature Research System Oct 2025 -- Jan 2026
- Designed a four-agent research pipeline (Planner, Retriever, Critic, Synthesizer) over Semantic Scholar's 200M+ paper index via direct REST integration with recency-weighted hybrid scoring; Critic issues PASS / STALE / CONTRADICTED / INSUFFICIENT verdicts and triggers targeted retries.
- Benchmarked against single-agent and naive multi-agent baselines on a 130-question eval built from real survey-paper ground truth: **52% staleness catch rate** vs. 0% for both baselines, with per-claim confidence and source attribution.

EmpathRAG -- Safety-Grounded Mental Health RAG Assistant Feb 2026 -- Apr 2026
- Built a 5-stage safety-gated NLP pipeline intercepting crisis messages via a DeBERTa NLI guardrail before the LLM runs -- 0/28 missed escalations vs. 9/28 unguarded baseline; **96.3% crisis recall**, 100% on euphemistic phrasing vs. 20% keyword filters.
- Implemented emotion-conditioned retrieval over 1.67M FAISS vectors, improving alignment 0.88 vs. 0.30 for a BM25 baseline; fine-tuned DeBERTa-v3 on 230K NLI pairs with a Captum audit trail on every intercept.

Publication: Voice-Driven Panoramic VR Generation -- IEEE IDCIoT 2024 (peer-reviewed, co-author). Speech-to-360° pipeline using Whisper, GPT-Neo, and Stable Diffusion; 2.3× throughput via attention slicing. DOI: 10467441